

Optimization Solution to the Multibody System of Anchor Chain and Comparison Analysis

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ABSTRACT

In order to describe the multibody system of the anchor chain with both the catenary and the ground chain under the environment of no special added equation or determinant condition, the optimization equation, i.e., the minimum potential energy equation, was proposed in this paper to represent the state of the static multibody system of anchor chain. This method was compared with the traditional catenary equation and the general mechanical equation of the multibody system of the anchor chain. The results showed that this method proposed in this paper need not the complex analysis involving forces. The form of the equation is simple and the physical meaning is clear and explicit. The computational results were consistent with the predictions from the traditional methods and this method in this paper can calculate the mixture of catenary and ground chain under the unified framework.

KEY WORDS: Anchor chain; Multibody system; Catenary equation; Optimization method; Ground chain; Principle of minimum potential energy.

INTRODUCTION

When the force analysis on anchor chain was carried out, it was reasonable to ignore the connection between the anchor rings and simplify the anchor chain as a catenary if the length of the anchor ring is small enough compared with the length of the overall anchor chain(Liu, Wang, 2011). The catenary equations derived from the differential equation of mechanics(Hu, Tang, 2004) could be applied to determine the parameters of the anchor chain and complete the selection of mooring equipment (Wang, Liu, 2007) in engineering. In addition, it was often used in the mooring of deep-water semi-submerged platform(Tong, Yang, et al., 2008) and FPSO (Floating Production Storage and Offloading)(Ting, Hong, et al., 2013), multi-component catenary mooring system (Zhu, Yin, et al., 2015), the intelligent active mooring system(Jamie, Simon, et al., 2016) and the design of mooring system(Zhu, Wang, et al., 2017).

There is no escape from occurrence of error in its physical model when simplifying the discrete multi-body system of anchor chain connected by anchor rings to a continuous catenary. Utilizing the multi-body dynamic equation to solve the state of the anchor chain system(Kreuzer,

Wilke, 2013) could not only eliminate the error of the physical model itself, but also be convenient for modeling and analyzing as a multi-body system(Choi, Sohn, 2013) together with the floating body.

Both the catenary equations and the multi-body system equations of anchor chain were derived based on Newton mechanics. The catenary equations with complex derivation process are a group of hyperbolic cosine equations with specific expressions, while the multi-body system equations of anchor chain are derived from the method of multi-body system (its basic principle is also based on Newton mechanics). In the multibody system method, the anchor chain is regarded as a system with multiple discrete and interconnected elements, which means it is more complex for deriving and solving. The optimization solution method proposed for anchor chain multibody system (the potential energy is minimum when the anchor chain is static) could be applied to solve the anchor chain multibody system with or without a ground chain in the unified framework, once establishing the optimization equations and combining the geometric constraints. There are a number of advantages in this approach including the unnecessary of complex force analysis, the easiness in building equations with universality and the applicability in solving static and dynamic problems.

ESTABLISHMENT AND SOLUTION OF MATHEMATICAL MODELS

Optimization expression of anchor chain multibody system

As shown in Fig. 1, assuming that the multibody system with uniform mass distribution consists of n rigid anchor rings connected head-to-tail and there is no friction between the rings. Since the rings are rigid, the bending moment, the torsion, and the shearing action of the rings are not considered. The initiating terminal number is 0, while the end point is numbered as n . S and H denote the horizontal and vertical projection of the anchor chain respectively as defined in Fig. 1. Let l_i denote the length of the i^{th} anchor ring, then the whole length of anchor chain is $\sum l_i = L$.

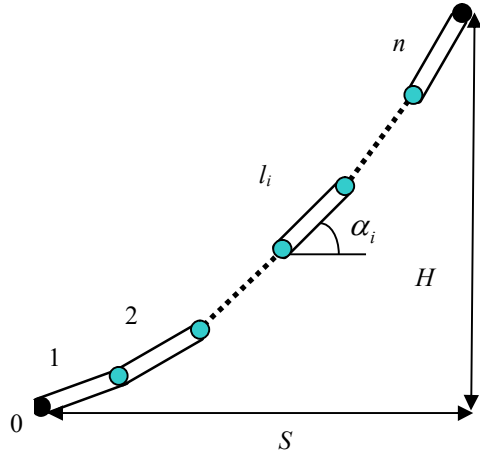


Fig. 1 Diagrammatic sketch to the multibody system of anchor chain

Assuming the both ends are fixed and defining the horizontal plane where the initial endpoint is located as the zero potential energy surface of the system, the anchor chain system is in static equilibrium under the action of gravity as depicted in Fig. 1. then P_i , the gravitational potential energy of the i^{th} anchor ring, can be written as:

$$P_i = m_i g \left[\sum_{j=1}^i l_j \sin(\alpha_j) - (l_i / 2) \sin(\alpha_i) \right] \quad (1)$$

Here, g is the gravitational acceleration; m_i is the mass of the i^{th} anchor ring; α_i denotes the angle subtended between the i^{th} anchor ring and the horizontal direction.

Since the anchor chain system is in a stable static state under the action of gravity, the total potential energy of the system reaches the minimum value at this moment, satisfying the following equation:

$$P = \min \left\{ \sum_{j=1}^n P_j \right\} \quad (2)$$

Both the points 0 and n are motionless, hence there are the following geometric relationships for the anchor chain system:

$$\begin{cases} \sum_{j=1}^n l_j \sin \alpha_j = H \\ \sum_{j=1}^n l_j \cos \alpha_j = S \end{cases} \quad (3)$$

The state of the anchor chain system is determined by the unknown variables α_i ($i=1,2,\dots,n$) which would be obtained through solving the Eq. (2) under the constraint Eq. (3). From the perspective of physics, Eq. (2) expresses the principle of minimum potential energy, that is to say, the total potential energy of the system is minimized when the system is in static status. Mathematically, however, it is a kind of problem on minimum with nonlinear constraints, which could be solved by Sequential Quadratic Program (SQP) algorithm. As one of the best algorithms for solving middle and small-scale nonlinear programming, the basic thought of SQP algorithm is transforming the nonlinear optimization problem with equality and inequality constraints into a quadratic programming problem. In each iteration a search direction of solution is determined through solving the quadratic programming subproblem and a merit function is used to get the moving step length. Repeating the iteration steps will finally find the solution of original problem (Ma, 2009).

Next, the calculation process of solving the above optimization issue on the basis of SQP algorithm is introduced in more details. First of all, the problem is expressed as the form of the nonlinear optimization problem as below:

$$\min f(x), \quad \text{subject to } c_i(x) = 0 \quad i=1,2 \quad (4)$$

Here, $f(x) = \sum_{j=1}^n P_j$ is the objective function;

$c(x) = \left[\sum_{j=1}^n l_j \sin \alpha_j - H, \sum_{j=1}^n l_j \cos \alpha_j - S \right]^T$ is the constraint function. The

Lagrangian function for this problem is expressed as:

$$L(x, \lambda) = f(x) - \lambda^T c(x) \quad (5)$$

Here, $\lambda = [\lambda_1, \lambda_2]^T$ is the Lagrangian multiplier vector. We use $A(x)$ to denote the Jacobian matrix of the constraints, that is,

$$A(x) = [\nabla c_1(x), \nabla c_2(x)]^T \quad (6)$$

Here, ∇ denotes gradient operation.

To define the subproblem the nonlinear constraints are linearized and Lagrangian function is approximated by quadratic polynomials to obtain:

$$\min \frac{1}{2} p^T H_k p + \nabla f(x_k)^T p \quad (7)$$

subject to $\nabla c_i(x_k)^T p + c_i(x_k) = 0 \quad i=1,2$

Here, H_k is the Hessian matrix of Lagrangian. To be practical, the quasi-Newton method is generally used to calculate B_k the approximation of Hessian matrix instead of Hessian matrix to overcome the problem of excessive computation of Hessian matrix. Hence, the subproblem is approximate to:

$$\min \frac{1}{2} p^T B_k p + \nabla f(x_k)^T p \quad (8)$$

subject to $\nabla c_i(x_k)^T p + c_i(x_k) = 0 \quad i=1,2$

Here, p_k defines the search direction in line search method during iteration process. The iteration is given by:

$$x_{k+1} = x_k + \psi_k p_k \quad (9)$$

Here, ψ_k is the step length deciding how far to move along the search direction p_k . To ensure that the SQP method converge from remoting starting points it is common to use a merit function to control the size of the steps (in line search methods) or to determine whether a step is acceptable. A variety of merit functions have been used in conjunction with SQP methods. Here one representative merit function of most those used in practice ℓ_1 is used with its usual definition of:

$$\ell_1(x; \mu) = f(x) + \frac{1}{\mu} \|c(x)\|_1 \quad (10)$$

Here, $\mu > 0$ is called the penalty parameter; $\| \cdot \|_1$ denotes the 1-norm operation. In iteration, the expression of calculating the Lagrangian multiplier is:

$$\lambda_{k+1} = [A_{k+1}^T A_{k+1}]^{-1} A_{k+1}^T \nabla f_{k+1} \quad (11)$$

The penalty parameter could be updated in accordance with:

$$\mu_k = \begin{cases} \mu_{k-1} & \mu_{k-1}^{-1} \geq \gamma + \delta \\ (\gamma + \delta)^{-1} & \mu_{k-1}^{-1} < \gamma + \delta \end{cases} \quad (12)$$

Here, $\gamma = \|\lambda_k\|_\infty$ is the infinite norm of λ_k ; $\delta > 0$ is a constant. The updating of Hessian matrix approximation is on the basis of the modified BFGS (Broyden, Fletcher, Goldfarb, Shanno) formula (Jorge, Stephen, et al., 2006) with its definition of:

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{r_k r_k^T}{s_k^T r_k} \quad (13)$$

Here, $s_k = \psi_k p_k$, r_k is defined as:

$$r_k = \theta_k y_k + (1 - \theta_k) B_k s_k \quad (14)$$

θ_k is defined in accordance with:

$$\theta_k = \begin{cases} 1 & s_k^T y_k \geq 0.2 s_k^T B_k s_k \\ \frac{0.8 s_k^T B_k s_k}{s_k^T B_k s_k - s_k^T y_k} & s_k^T y_k < 0.2 s_k^T B_k s_k \end{cases} \quad (15)$$

The expression of y_k is:

$$y_k = \nabla_x L(x_{k+1}, \lambda_{k+1}) - \nabla_x L(x_k, \lambda_k) \quad (16)$$

The analysis above established the mathematical model of the anchor chain multibody system in an optimization expression, it could be solved by SQP approach. Nevertheless, how to solve the corresponding quadratic programming is not pointed out, next it will be described in details.

In order to facilitate numerical calculation, the quadratic programming subproblem described in Eq. (8) can be equivalent to the following Eqs. (17) and (18) (Ma, 2009):

$$F_1(p, \lambda) = B_k p - A_k^T \lambda + \nabla f(x_k) = 0 \quad (17)$$

$$F_2(p, \lambda) = c(x_k) + A_k^T p = 0 \quad (18)$$

Set $z = (\xi, p, \lambda)$, then the Eqs. (17) and (18) could be merged into:

$$F(z) = F(\xi, p, \lambda) = \begin{bmatrix} \xi \\ F_1(p, \lambda) \\ F_2(p, \lambda) \end{bmatrix} = 0 \quad (19)$$

The Jacobian matrix of $F(z)$ is written as:

$$J_F(z) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & B_k & -A_k^T \\ 0 & A_k & 0 \end{bmatrix} \quad (20)$$

Set a parameter satisfying $\eta \in (0, 1)$, then the non-negative function is defined as:

$$\beta(z) = \eta \|F(z)\| \min \{1, \|F(z)\|\} \quad (21)$$

The numerical simulation steps for solving the SQP subproblem are listed as below:

Step 1: choose parameters $\tau, \sigma, \eta, \xi_0 \in (0, 1)$, $\varepsilon_0 > 0$ (the specific values are $\tau=0.5$, $\sigma=0.2$, $\eta=0.05$, $\xi_0=0.05$, allowable tolerance $\varepsilon_0=1.0 \times 10^{-6}$) to satisfy the following relationships $\eta \xi_0 < 1$ and $\eta \|F(z_0)\| < 1$; set $z_0 = (\xi_0, p_0, \lambda_0)$, $\bar{z} = (\xi_0, 0, 0)$, $j = 0$.

Step 2: judge whether the relationship $\|F(z_j)\| \leq \varepsilon_0$ is satisfied. If yes, stop; otherwise compute $\beta_j = \beta(z_j)$ according to formula (21).

Step 3: solve the following equation to obtain the solutions $\Delta z_j = (\Delta \xi_j, \Delta p_j, \Delta \lambda_j)$.

$$F(z_j) + J_{F(z_j)} \Delta z_j = \beta_j \bar{z}$$

Step 4: assume that b_j is the minimum non-negative integer satisfying the following inequality. Once b_j is obtained, set $\psi_j = \tau^{b_j}$ and $z_{j+1} = z_j + \psi_j \Delta z_j$.

$$\|F(z_j + \tau^{b_j} \Delta z_j)\| \leq [1 - \sigma(1 - \eta \xi_0) \tau^{b_j}] \|F(z_j)\|$$

Step 5: set $j = j + 1$ and go to step 2.

Based on the solving method and steps of the SQP subproblem above, the computational process of the SQP algorithm for the proposed problem in the paper is presented as following.

Step 1: choose parameters $\eta \in (0, 0.5)$, $\tau \in (0, 1)$, allowable tolerance $0 \leq \varepsilon_1, \varepsilon_2 \leq 1$, $\delta > 0$ (the specific values are $\eta=0.1$, $\tau=0.5$, $\varepsilon_1=1.0 \times 10^{-6}$, $\varepsilon_2=1.0 \times 10^{-5}$, $\delta=0.05$); set $k = 0$; choose initial values: B_0 is a unit matrix with the size of $n \times n$, $\mu_0=0.8$, input the initial values x_0, λ_0 ; evaluate $f_0, \nabla f_0, c_0, A_0$.

Step 2: compute p_k by solving the quadratic programming subproblem, namely formula (8).

Step 3: judge whether the inequalities $\|p_k\|_1 \leq \varepsilon_1$ and $\|c_k\| \leq \varepsilon_2$ are satisfied. If so, stop the computation and the approximation x_k for original problem is found.

Step 4: update the penalty parameter μ_k in accordance with formula (12).

Step 5: Armijo search. Determine the minimum non-negative integer b_k satisfying the following inequality:

$$\phi_1(x_k + \tau^{b_k} p_k; \mu_k) - \phi_1(x_k; \mu_k) \leq \eta \tau^{b_k} \phi_1'(x_k; p_k)$$

Set $\psi_k = \tau^{b_k}$.

Step 6: obtain x_{k+1} by updating x_k through formula (9), then evaluate $f_{k+1}, \nabla f_{k+1}, c_{k+1}, A_{k+1}$. Update B_{k+1} according to Eqs. (13)~(16).

Step 7: set $k = k + 1$ and go to step 2.

It is worth mentioning that when the anchor chain is long enough the ground chain will exist. It is convenient to determine the state of the system with or without the ground chain by defining the lower boundary of α_i , whereas, other methods require a cyclic judgment or restriction equations.

Since the points 0 and 1 of the first anchor ring are in the same line, it is reasonable to assume that $\alpha_0 = \alpha_1$. Let T_i ($i = 0, 1, \dots, n$) denote the force on the point i , then the static equilibrium of the system implies:

$$\begin{cases} T_0 \cos \alpha_0 = T_n \cos \alpha_n \\ T_0 \sin \alpha_0 + \sum_{j=1}^n m_j g = T_n \sin \alpha_n \end{cases} \quad (22)$$

Application of relevant transformations means that:

$$\begin{cases} T_0 = \frac{\left(\sum_{j=1}^n m_j g \right) \cos \alpha_n}{\cos \alpha_0 \sin \alpha_n - \cos \alpha_n \sin \alpha_0} \\ T_n = \frac{\left(\sum_{j=1}^n m_j g \right) \cos \alpha_0}{\cos \alpha_0 \sin \alpha_n - \cos \alpha_n \sin \alpha_0} \end{cases} \quad (23)$$

As can be seen from the Eq. (23), the anchor chain force (catenary part) is only related to three physical quantities: the angle of start point α_0 , the angle of end point α_n and the total mass of anchor chain $\sum_{j=1}^n m_j g$.

General mechanics equations of the multibody system of the anchor chain

As depicted in Fig. 2, F_{xi} and F_{yi} are respectively the horizontal and vertical components of the force on the start point of the i^{th} anchor ring, while $F_{x(i+1)}$ and $F_{y(i+1)}$ denote the corresponding components on the end point respectively. The equilibrium of the system leads to the relationships:

$$\begin{cases} F_{xi} = F_{x(i+1)}, & (i = 1, 2, 3, \dots, n) \\ F_{yi} + m_i g = F_{y(i+1)}, & (i = 1, 2, 3, \dots, n) \end{cases} \quad (24)$$

Here, F_{xi} and F_{yi} ($i = 1, 2, 3, \dots, n+1$) are variables to be determined, which means the number of undetermined variables is $2n+2$. Since within Eq. (24) the number of unknowns equals $2n$, the constraint conditions need to be added. That is:

$$\begin{cases} \sum_{i=1}^n l_i \sin \alpha_i = H \\ \sum_{i=1}^n l_i \cos \alpha_i = S \end{cases} \quad (25)$$

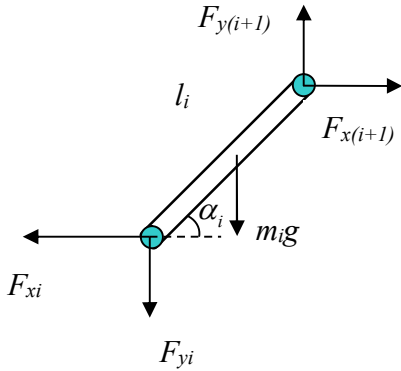


Fig. 2 Diagrammatic sketch for force analysis of single anchor ring

Due to the moment equilibrium for the i^{th} anchor ring, it can be established that:

$$F_{yi} l_i \cos \alpha_i + m_i g \frac{l_i}{2} \cos \alpha_i = F_{xi} l_i \sin \alpha_i \quad (26)$$

Application of relevant transformation leads to:

$$\tan \alpha_i = \frac{\sin \alpha_i}{\cos \alpha_i} = \frac{F_{yi} l_i + m_i g \frac{l_i}{2}}{F_{xi} l_i}$$

Namely:

$$\alpha_i = \arctan \left(\frac{\sin \alpha_i}{\cos \alpha_i} \right) = \arctan \left(\frac{F_{yi} l_i + m_i g \frac{l_i}{2}}{F_{xi} l_i} \right) \quad (27)$$

Hence F_{xi} , F_{yi} and α_i are immediately determinable from Eqs. (24), (25) and (27).

Derivation of Catenary equations

Taking the anchor chain as an inelastic line with uniform mass distribution and assuming that the gravity of unit length is q then for the infinitesimal amount ds at any position of the anchor chain its gravity can be written as $dG = qds$. The angle between the tangential direction and horizontal direction at the start point is denoted as θ and the corresponding angle at other point is $(\theta + d\theta)$, hence the horizontal force on the arbitrary point T_h satisfies:

$$T(x) \cos \theta = T(x + dx) \cos(\theta + d\theta) \equiv T_h \quad (28)$$

Due to

$$\begin{aligned} ds &= \frac{dx}{\cos \theta} = \sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta}} dx \\ &= \sqrt{1 + \frac{\sin^2 \theta}{\cos^2 \theta}} dx = \sqrt{1 + \tan^2 \theta} dx \\ &= \sqrt{1 + y'^2} dx \end{aligned}$$

the force of the arbitrary point in vertical direction can be expressed as:

$$T(x) \sin \theta = T(x + dx) \sin(\theta + d\theta) - q \sqrt{1 + y'^2} dx \quad (29)$$

It can be deduced by combining Eqs. (28) and (29) that:

$$T_h \tan(\theta + d\theta) - T_h \tan \theta - q \sqrt{1 + y'^2} dx = 0 \quad (30)$$

namely

$$T_h dy' - q \sqrt{1 + y'^2} dx = 0 \quad (31)$$

Application of relevant transformation implies that:

$$\frac{dy'}{\sqrt{1 + y'^2}} = \frac{dx}{a} \quad (32)$$

with $a = T_h / q$. Rewriting Eq. (32) as the second order ordinary differential equation form leads to:

$$y'' = \frac{1}{a} \sqrt{1 + y'^2} \quad (33)$$

The general solution of Eq. (33) is:

$$y = C_1 + a \cosh \left(\frac{x}{a} + C_2 \right) \quad (34)$$

The solution of the differential equation can be determined with the help of boundary conditions. Under the condition of no ground chain, the boundary conditions are given by:

$$\begin{cases} y(0) = 0 \\ y'(0) = \tan \theta \end{cases}$$

Consequently, the particular solution is

$$y = a \cosh \left(\frac{x}{a} + \ln(\tan \theta + \sec \theta) \right) - a \sec \theta \quad (35)$$

Since $y(S) = H$, it follows that:

$$H = a \cosh \left(\frac{S}{a} + \ln(\tan \theta + \sec \theta) \right) - a \sec \theta \quad (36)$$

Due to $ds = \sqrt{1 + y'^2} dx$, thus the length from the arbitrary point to the origin satisfies:

$$l = \int ds = a \sinh \left| \frac{x}{a} + \ln(\tan \theta + \sec \theta) \right| - a \tan \theta \quad (37)$$

Therefore, the total length of the anchor chain is expressible as:

$$L = \int ds = a \sinh \left| \frac{S}{a} + \ln(\tan \theta + \sec \theta) \right| - a \tan \theta \quad (38)$$

The water depth H , the horizontal projection S of anchor chain and the overall length L of anchor chain are generally given, hence the variables θ and a can be determined through combining Eqs. (36) and (38). Substituting θ and a into Eq. (35) could establish the specific expression of the catenary equation, thereby the tangential direction can be solved. Furthermore, the anchor chain force at the initial and end points also can be determined in accordance with the tangential angles at both ends and the total gravity of anchor chain.

EXAMPLES AND COMPARISON ANALYSIS

In order to establish the correctness of the optimization solution method developed, the predictions based on the algorithmic approaches mentioned above are compared with the corresponding results generated by literatures(Hou, Dong, et al., 2005; Zheng, 2006). Relevant parameters and results are summarized in Table 1.

Table 1 Calculation results and corresponding comparisons

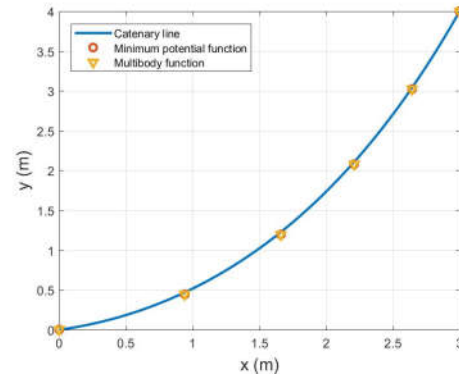
Case	Anchor chain length (m)	Horizontal projection of anchor chain (m)	Vertical projection of anchor chain (m)	Mass per unit length ($\text{kg} \cdot \text{m}^{-1}$)	Horizontal force of anchor chain (N)		Error (%)
					Reference method	Principle of minimum potential energy	
1	20	11	15	25	127.334	131.365	3.17
2	114.4	100	48.67	219.16	235900	236180	0.12

It can be found from Table 1 that the predictions based on the principle of minimum potential energy proposed in the paper are very close to the results from literatures(Hou, Dong, et al., 2005; Zheng, 2006) and the errors are less than 4%, demonstrating the practicability and effectiveness of the optimization method.

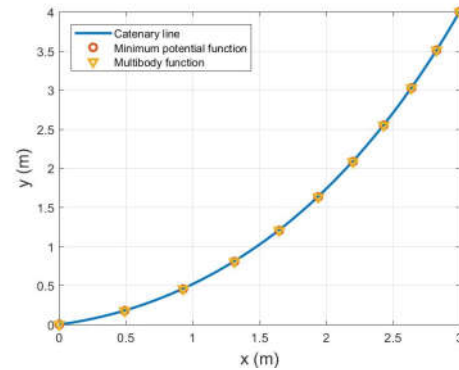
Next, the comparison analysis with regard to the numerical results obtained by the principle of minimum potential energy, general mechanics method of anchor chain multibody system and the catenary equations is carried out. The parameter values of the calculation model are listed as below: the horizontal projection of the anchor chain $S=3.0\text{m}$, the vertical projection of the anchor chain $H=4.0\text{m}$, the mass per unit length $\bar{m}=10.0\text{kg/m}$, the gravitational acceleration $g=9.81\text{m} \cdot \text{s}^{-2}$. In accordance with the derivation and solving methods developed, once the anchor chain length L and the number n of anchor ring are given, the system state could be determined by solving Eq. (2) with the constraint Eq. (3) as the optimization method described. Combining Eqs. (24), (25) and (27) could determine the anchor chain multibody system state as well, according to the general mechanics method. Through combining catenary Eqs. (36) and (38) the state of the system is determinable if the anchor chain length L is known.

Set $L=5.2\text{m}$ and $n=5, 10, 20$ respectively, thus the comparisons of corresponding results are depicted in Fig. 3. The legend of solid line stands for the results generated by the catenary equations; the legend of hollow point illustrates the predictions based on the optimization approach; the results from the general mechanics multi-body system method are denoted by the inverted triangles. As demonstrated in Fig. 3(a) there are some differences between the predictions from the multibody system methods and the catenary equations under the condition of $n=5$. This is because the single segment is not a small quantity compared with the whole length of the anchor chain.

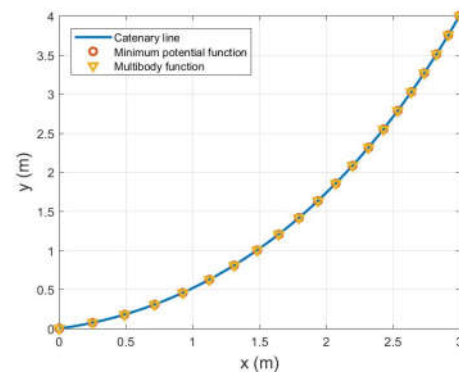
From Fig. 3(b) and (c), namely when $n=10$ and 20 respectively, the corresponding nodes fit well with the catenary. It can be seen from Fig. 3(c) that the predictions based on three different methods are almost consistent with each other and then verified the effectiveness and reliability of the optimization method.



(a) $n=5$



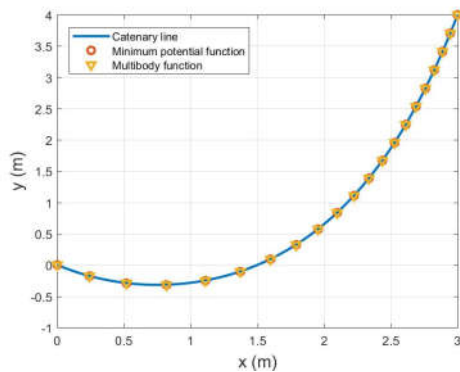
(b) $n=10$



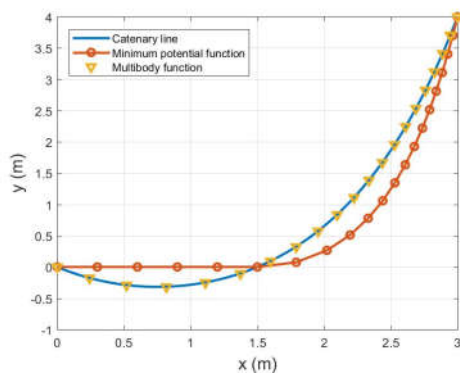
(c) $n=20$

Fig.3 The results of various methods with different element nodes ($L=5.2\text{m}$)

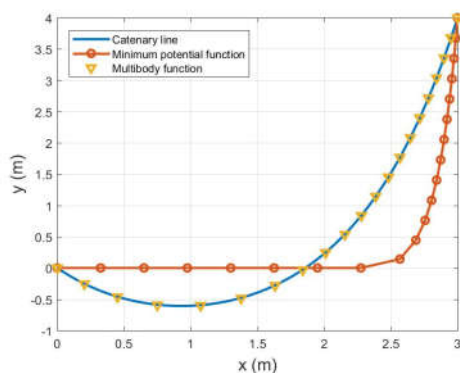
Next, increasing the length of anchor chain to $L = 6.0\text{m}$ and setting $n = 20$, the results are shown in Fig. 4(a). The corresponding conclusion is consistent with what Fig. 3 reveals. For the condition with ground chain, additional equations or judgements are needed for the general mechanics method of multibody system and catenary method, for instance assuming that the initial point of the anchor chain system is located at the intersection of horizontal plane and the catenary(Wang, Liu, 2007). As demonstrated in Fig. 4(b) and (c), however, the state with or without ground chain is convenient to be determined though the optimization method just by defining the lower boundary values of α_i , such as $\alpha_i \geq 0$. Consequently, it is impossible for the other two methods to solve in a unified framework, whereas the optimization approach does.



(a) without ground chain ($L = 6.0\text{m}$)



(b) with ground chain ($L = 6.0\text{m}$)



(c) with ground chain ($L = 6.5\text{m}$)

Fig.4 Comparisons among different methods of various lengths with or without ground chain

CONCLUSIONS

An optimization solution method (principle of minimum potential energy) with simple derivation, clear and explicit physical meaning and simplicity in solving is proposed in this paper to determine the static state of anchor chain multibody system. Through comparing with the results obtained by catenary equations and general mechanics equations of anchor chain multibody system, it is found that there are some differences between the predictions from continuous catenary equations and multibody system methods when the anchor ring length has the same order with the length of whole anchor chain. However, if the anchor ring length is less than the length of whole anchor chain with more than one order, the differences almost disappear. It can be seen from the relative results that the other two methods commonly applied are difficult to be solved under a unified framework. They have complex steps and hence need to be improved, while the optimization approach based on the principle of minimum potential energy proposed in the paper could overcome the shortcomings successfully and therefore it has great value in engineering application.

ACKNOWLEDGEMENTS

This project is supported by the National Natural Science Foundation of China(Grant No. 51879058), the Natural Science Foundation of Jiangsu Province(Grant No. BK20161468) and Qing Lan Project.

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